PROPERTIES $\TAU$ LOCAL ON THE SOURCE ARE STABLE UNDER PRECOMPOSITION WITH $\TAU$-FLAT MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Load-bearing glue lemma: the delivered card for lemma-etale-smooth-local-source-target (L30063) invokes this lemma in its first step.

Allowed standard prerequisites (editorial).

- Definition of τ local on the source. Stacks Project, Descent on Spaces, Definition definition-property-morphisms-local-source. Supplies the locality being applied to the assembled covering.
- τ-coverings for $\tau \in \{fpqc, fppf, syntomic, smooth, \etale\}$. Stacks Project, Topologies on Spaces, Section on coverings (spaces-topologies-lemma-fpqc and siblings). Supplies the covering classes into which the given morphism is fitted.

Lemma 0.1. *Let S be a scheme. Let $\tau \in \{fpqc, fppf, syntomic, smooth, étale\}$. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces over S which is τ local on the source. Let $f : X \to Y$ have property $\mathcal{P}$. For any morphism $a : X' \to X$ which is flat, resp. flat and locally of finite presentation, resp. syntomic, resp. smooth, resp. étale, the composition $f \circ a : X' \to Y$ has property $\mathcal{P}$.*

Proof. This is true because we can fit $X' \to X$ into a family of morphisms which forms a τ-covering. □

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